
Where Does Perceptual Compression Live? An Earned Null for Cone-Organized Diminishing Returns

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Abstract

Diminishing returns (DR) in color dissimilarity challenges additive geodesic-distance models, but its channel organization is unknown. We test whether compression separates along L , M , S cone axes using public Helm and Ekman judgments converted to a stated normalized proxy for CIE 2006 LMS. A gauge-fixed nested family compares additive M0, isotropic-DR M1, cone-axis M2, and matched M3 models on opponent, sRGB, CIE RGB, CIELAB, and LCh/HCL coordinates. Pair-disjoint prediction with 200 paired training-cluster bootstraps shows that M1 improves over M0 by +0.453 nats/response (95% CI [0.246,0.524]), but M2 does not improve over M1 (-0.035, [-0.102,0.050]) and loses to the resample-selected best M3 (-0.258, [-0.387,-0.132]). The 2022 gray DR result replicates independently. Observed-design simulations distinguish cone from opponent generators with 98% to 100% accuracy, yet individual cone exponents recover poorly; a powered coarse LM/S comparison also fails. Direct-chord, path, spectral, coordinate-quality, and function-family controls preserve the verdict. These archives support perceptual compression and some non-cone axis dependence, but neither a full nor coarse cone organization; they do not support locating DR before cone-signal combination.

1 Introduction

A Riemannian model turns local color discrimination into global distance by integrating infinitesimal steps. If A, B, C occur in order on a geodesic, that construction requires $D(A, B) + D(B, C) = D(A, C)$. Human judgments on the neutral lightness axis instead show diminishing returns (DR): the direct large difference is smaller than its parts [Bujack et al., 2022]. Conditional on the neutral locus being a perceptual geodesic, this is not merely a poor Euclidean embedding; it contradicts additivity of Riemannian geodesic length.

The next question is where that compression lives. Two chromatic-line experiments established DR and showed that one line’s scaling does not transfer unchanged to another [Stark et al., 2025]. Direction dependence by itself is therefore not new. The open proposal is more specific: perhaps each cone difference is compressed before cone signals are combined,

$$D(\Delta c) = F(f_L(|\Delta L|) + f_M(|\Delta M|) + f_S(|\Delta S|)).$$

That hypothesis could replace an atlas of direction-specific corrections with three inner functions and one outer scaling. It also has a biological reading: alignment with L, M, S would place compression before channel combination, whereas alignment with $L + M, L - M$, or other derived coordinates would place the useful nonlinearity after aggregation. Conversely, if equally flexible standard non-cone bases match or beat LMS, a cone-specific mechanism and its compact whole-space prediction are not established.

Testing that claim is harder than fitting three flexible curves. LMS and opponent coordinates are nearby rotations, archival stimuli cover directions unevenly, and L and M differences co-vary. Extra

parameters can only count through held-out prediction. Moreover, hue stimuli lie on curved loci: subadditivity of accumulated hue steps can arise from chord–arc geometry even under an additive ambient metric. A credible result therefore needs a matched axis panel, an explicit scale gauge, recovery and nearest-control power analysis, and a curvature baseline.

We conduct that test using only the supplied public human judgments: Helm’s surface-color ratio dissimilarities, Ekman’s spectral-color aggregate matrix, the 2022 gray triads, and Wiebel et al.’s matching/MLDS data. Stimuli are placed in a stated von-Kries-normalized proxy for CIE 2006 LMS using the two supplied fundamental tables, with the unavoidable observer-bridge approximation fully logged. We compare additive M0, isotropic-DR M1, cone-axis M2, and five matched-complexity M3 controls (opponent, sRGB, CIE RGB, CIELAB, and CIE LCh/HCL) under pair-disjoint cross-validation and selection-aware bootstrap uncertainty.

Our contributions are:

1. a gauge-fixed, genuinely nested and cross-validated cone-alignment test against the complete fixed-basis panel, rather than a claim from anisotropy alone;
2. an independent gray replication plus a matching-to-MLDS suprathreshold check, anchoring the observation pipeline before chromatic extension;
3. identifiability evidence that separates model-selection power from per-cone parameter recovery and includes a powered coarse LM/S-versus-opponent test; and
4. an explicit open/periodic hue-locus arc–chord analysis, along with polygon, smooth-path, local-pair, drop-C, and exact-spectral sensitivities.

The resulting answer is an earned null for cone organization within this model family. Isotropic compression strongly improves chromatic prediction (+0.453, 95% CI [0.246,0.524]), but cone-axis flexibility does not improve on it (-0.035, [-0.102,0.050]) and loses to the resample-selected non-cone panel by -0.258 ([-0.387,-0.132]). The best point control is CIE LCh/HCL. Synthetic cone-versus-opponent selection is powered in both generating directions, while the coarse LM/S model also fails on real held-out data. Thus these public data support robust compression but neither a three-cone decomposition nor a pre-combination cone mechanism.

2 Related Work

Bujack et al. demonstrated concave difference scaling on the neutral L^* axis using Thurstonian triads [Bujack et al., 2022, Teti et al., 2022]. Conditional on that locus being a perceptual geodesic, strict subadditivity is incompatible with Riemannian geodesic distance. Our gray analysis is an independent pipeline anchor for that result, not a new discovery.

Stark, Turton, and Bujack extended the triad method to two chromatic CIELAB lines and found DR on both, while a function trained on one line transferred poorly to the other [Stark et al., 2025]. They explicitly proposed a per-cone decomposition and stated that two geodesics could not test it. Consequently, chromatic DR, non-constancy, and mere anisotropy are already prior results. The new falsifier here is alignment: a cone-separable function must beat equally flexible functions on several fixed standard bases, especially the nearby opponent basis, on unseen pairs.

The companion geometric work formalizes hue, saturation, lightness, and induced path metrics in a non-Riemannian setting [Bujack et al., 2025]. Its experiments did not find bias-corrected evidence that shortest stimulus paths and induced-metric paths diverged; it did not fit LMS channel functions or compare axis bases. Path-locus geometry and the alignment of distance compression are therefore complementary questions.

Historical multidimensional scaling already showed that no single Euclidean arrangement captures every color-dissimilarity matrix [Ekman, 1954, Helm, 1964]. Standard CIELAB and modern color-difference formulas were designed primarily for local or industrial prediction [Luo et al., 2001, International Organization for Standardization and Commission Internationale de l’Eclairage, 2022]; their success as controls here is evidence about predictive coordinates, not proof that their axes are neural channels. Likewise, our fixed opponent coordinates are a matched alternative motivated by post-receptor combination [MacLeod and Boynton, 1979], not a uniquely canonical physiological transform.

3 Methodology

3.1 Question, estimand, and decision rule

We separate three questions that are easily conflated. M1 versus M0 asks whether a direction-independent compression improves prediction over an additive distance. M2 versus M1 asks whether allowing distinct functions on the cone axes helps out of sample. Finally, M2 versus a matched panel M3 asks whether any such structure is *specifically* cone-aligned. Our primary score is held-out Gaussian log predictive density per response (nats; larger is better), first averaged within Helm and Ekman and then equally across the two datasets. Equal dataset weighting prevents Helm’s session-level replication from overwhelming Ekman’s aggregate matrix. Normalized RMSE is secondary. Cone specificity requires the 95% interval for

$$\min_{k \in \mathcal{K}} \{\text{score}(\text{M2}) - \text{score}(\text{M3}_k)\}$$

to exceed zero, where $\mathcal{K}$ contains all five controls and the minimizing control is reselected in every bootstrap draw. This is a deliberately harder, selection-aware standard than choosing one control on the complete sample.

3.2 From supplied colorimetry to normalized cone coordinates

The stimulus tables provide CIE 1931 xyY . We convert $X = xY/y$ and $Z = (1 - x - y)Y/y$ [Commission Internationale de l’Eclairage, 2018]. The accompanying cone bundle contains the CIE 1931 2-degree XYZ color-matching functions and energy-basis CIE 2006 2-degree LMS fundamentals [Commission Internationale de l’Eclairage, 2019, 2006], but—contrary to the bundle description in the assignment—no LMS-to-XYZ matrix or provenance file. Consequently, we align the two supplied tables at each integer wavelength from 390 to 830 nm (blank S entries above 615 nm are physical zeros) and estimate one response-blind least-squares bridge

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} = \begin{bmatrix} 0.268274 & 0.846614 & -0.034892 \\ -0.385945 & 1.164877 & 0.102851 \\ 0.021247 & -0.024552 & 0.534161 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}.$$

This is an approximate projection between different standard observers, not an official CIE physical transform. We therefore log channelwise residuals and run a direct fundamental-lookup sensitivity for Ekman’s monochromatic lights; the analogous validation is impossible for Helm without chip spectra and an Illuminant-C spectral power distribution.

Both datasets use CIE Illuminant C, $(x_w, y_w) = (0.31006, 0.31616)$, as the stated nominal reference. Each projected component is divided by the corresponding projected white component, giving von-Kries coordinates with white $(1, 1, 1)$. This algebraic white check and the round trip through the inverse matrix are supplemented by logged conversions of Helm A and the 555-nm Ekman stimulus. Every stimulus coordinate is retained in `stimulus_coordinates.csv`.

The fixed controls are: (i) an orthonormal opponent transform

$$\left((L - M)/\sqrt{2}, (2S - L - M)/\sqrt{6}, (L + M)/\sqrt{2} \right);$$

(ii) linear sRGB after a fixed Bradford C-to-D65 adaptation, without clipping [Commission Internationale de l’Eclairage, 2004, International Electrotechnical Commission, 1999]; (iii) fixed CIE RGB; (iv) CIELAB relative to C; and (v) cylindrical CIELAB LCh/HCL [International Organization for Standardization and Commission Internationale de l’Eclairage, 2019]. For the last control we use the standard signed pair component $\Delta H^* = 2\sqrt{C_i^* C_j^*} \sin(\text{wrap}(h_j - h_i)/2)$ alongside ΔL^* and ΔC^* . Thus Lab and LCh are evaluated in their own fixed nonlinear coordinates; neither uses a tuned tangent frame. Before fitting, every axis of every basis is divided by its RMS difference over all stimulus pairs in that dataset. This diagonal, response-blind conditioning does not rotate axes. No transform is fitted to judgments.

3.3 Gauge-fixed nested distance family

For an absolute RMS-normalized coordinate difference $x \geq 0$, define

$$\phi_p(x) = \frac{(1+x)^p - 1}{2^p - 1}, \quad F_q(z) = \frac{(1+z)^q - 1}{q},$$

with continuous limits $\phi_0(x) = \log(1+x)/\log 2$ and $F_0(z) = \log(1+z)$. We constrain $p, q \in [-5, 1]$, for which these functions are increasing and concave through the linear endpoint. The gauge is explicit: $\phi_p(0) = 0$, $\phi_p(1) = 1$, $F_q(0) = 0$, $F_q'(0) = 1$, positive weights α_k sum to one, and observation links absorb dataset units. Thus the outer curvature cannot be arbitrarily traded against inner scale.

On LMS differences $x_c = |\Delta c|$, the nested models are

$$D_{M0} = \sum_c \alpha_c x_c, \quad (1)$$

$$D_{M1} = F_q \left(\sum_c \alpha_c x_c \right), \quad (2)$$

$$D_{M2} = F_q \left(\sum_c \alpha_c \phi_{p_c}(x_c) \right). \quad (3)$$

M0 is recovered from M1 at $q = 1$, and M1 from M2 at $p_L = p_M = p_S = 1$. Each M3 substitutes one fixed control basis for LMS and has the same two free weight logits, outer parameter, and three inner parameters as M2. A prespecified anchored-log family repeats the entire panel. All optimizations use bounded L-BFGS-B with deterministic multistarts; objectives, statuses, boundary hits, and seeds are logged.

M0 uses city-block aggregation and is an additive no-compression null, not a Riemannian L_2 metric. Our target is the location of compression within this nested additive-aggregation family. The L_1 versus L_2 choice is outside the present identification claim.

3.4 Observation models and held-out prediction

Helm’s ratio dissimilarities follow a zero-intercept Gaussian link $y_{sp} \sim \mathcal{N}(b_s D_p, \sigma_{\text{Helm}}^2)$, with a positive training slope b_s for each session. N6a and N6b retain separate slopes but share an observer bootstrap cluster. Ekman’s aggregate dissimilarities use $y_p \sim \mathcal{N}(b_E D_p, \sigma_{\text{Ekman}}^2)$. For a candidate latent distance the positive slope has its profiled training least-squares solution; each dataset’s residual scale is also estimated only on the training fold. The shared object is the distance-function form and its parameters, not a naively pooled response scale.

Five folds are separation-stratified and unordered-pair-disjoint: all Helm sessions for one pair remain together. Point estimates concatenate all out-of-fold predictions. In each of 200 paired bootstrap replicates, training pairs are resampled within dataset and Helm observers are resampled with N6’s sessions grouped; the held-out pairs never enter fitting. Every model sees the same draw. Percentile intervals are paired across models. The selected best M3 is recomputed within each draw.

3.5 Gray replication and independent suprathreshold check

The gray release lacks participant identifiers, so we aggregate the 91,896 binary responses into exact binomial counts and keep all five standards for one of the 64 signed offset configurations in the same fold. Both gray models fit a monotone piecewise-linear nuisance map $g(L^*)$ with knots 0, 25, 50, 75, 100, endpoints 0 and 1, and four positive simplex increments. Every gray stimulus varies only on the neutral L^* coordinate, so every retained triple is exactly collinear and passes the prespecified near-collinear-triple restriction. Under the supplied response coding,

$$P(R = 1) = \Phi(\beta [F_q(|g(L_{t2}) - g(L_s)|) - F_q(|g(L_{t1}) - g(L_s)|)]).$$

M0 fixes $q = 1$; M1 estimates q . We report held-out binomial log score and the learned- g ratio of a direct 30-unit difference to its two adjacent 15-unit parts, averaged over all valid 2.5-unit origins. This scale-free magnitude is comparable to the concrete ratio reported in 2022.

As an independent suprathreshold check, Wiebel et al.’s plain-context matching responses first estimate each observer’s ten-point lightness map without using MLDS choices [Wiebel et al., 2017]. M0 and M1 then predict held-out observers’ plain-context MLDS choices through a Thurstonian probit link [Thurstone, 1927, Maloney and Yang, 2003]. These choices are not pooled with the 2022 task.

3.6 Identifiability and curvature controls

We compute signed and absolute LMS correlations, VIFs, standardized singular values, condition numbers, and direction-covariance eigenvalues from unique pairs. Synthetic outcomes preserve the exact directions, folds, session links, and fitted noise. Two hundred full-M2 simulations measure recovery of (p_L, p_M, p_S) . Two hundred datasets generated under M2 and 200 under the nearest fixed opponent M3 are refit by both models; bidirectional correct selection with a Wilson interval quantifies whether alignment is resolvable.

The prespecified coarse comparison uses $(|\Delta L| + |\Delta M|, |\Delta S|)$ versus a matched derived-channel control $(|\Delta(L - M)|, |\Delta(S - (L + M)/2)|)$, each reconditioned by response-blind axis RMS. The coarse model must beat both its isotropic nested version and the matched opponent control on held-out data, with corresponding bidirectional simulation power, before we call even a coarse channel split identified.

For curvature, Helm follows the fixed periodic order A,C,E,G,I,K,M,O,Q,S; Ekman is an open wavelength arc and is never closed by an invented purple edge. We compare accumulated polygonal arc exposure with the direct endpoint chord under linear and concave scalar models. The full panel is repeated on per-axis accumulated exposure, on response-blind periodic/natural cubic exposures, and on low-curvature pairs (separation at most two and chord/arc at least 0.90). A drop-C panel addresses Helm C’s internally inconsistent hue coordinate without guessing a replacement.

4 Experimental Setup

Human data. The 2022 gray release contains 91,896 retained choices over 320 triads (five standards times 64 signed offset configurations); its repeated copies in the workspace are byte-equivalent and counted once. The accompanying LANL repository is BSD-3-Clause [Los Alamos National Laboratory, 2022]. Wiebel et al. provide 10 observers, ten reflectance levels, five contexts, 10 repeated matches per level/context, and 10 repeats of every MLDS triad; the supplied archive states CC BY 4.0 [Wiebel et al., 2017, 2022]. We use the 1,000 plain-context matching trials to define observer scales and the 12,000 plain-context MLDS choices for the independent check.

Helm contributes all 45 pairs of ten Illuminant-C Value-6 tiles. The main analysis uses 495 responses: 11 normal-session matrices from ten people, with N6a/N6b recognized as repeat sessions. The authoritative long file is used; the convenience average silently omits N10 and counts N6 twice. The separate five color-deficient sessions are a sensitivity. Ekman contributes the upper triangle of a symmetric 14-by-14 aggregate matrix (91 pairs; approximately 31 observers) over 434–674 nm spectral lights. Both matrices are redistributed through `smacof` under GPL-3 and retain their original scientific attribution [Helm, 1964, Ekman, 1954, Mair et al., 2022].

Checks and preprocessing. The conversion script records SHA-256 hashes, the full projection and inverse, all reference-white and chromatic checks, and every coordinate. Helm C and E are only $\Delta E_{ab}^* = 2.922$ apart under the supplied coordinates despite substantial judged separation; C is therefore retained in the contract-fixed primary analysis and removed, without replacement, in a declared sensitivity. Ekman’s bounded dissimilarities have three off-diagonal values at 1.0; the Gaussian link is consequently understood as a predictive working likelihood, not a claim of unbounded response generation.

Folds, uncertainty, and compute. Helm has exactly nine pairs in every fold; Ekman has 19 in one and 18 in each other fold. Gray configuration folds contain 60 or 65 triads. The fixed seed is 20260728. Primary intervals use 200 training-cluster bootstrap refits; identifiability uses 200 simulations per generating condition. NumPy 2.5.1, SciPy 1.18.0, and Matplotlib 3.11.1 are vendored locally. The complete analysis ran on CPU, with independent bootstrap/simulation replicates parallelized and all random seeds and convergence statuses retained.

Gray anchor. Before chromatic extension, held-out M1 beats M0 by +0.0083 nats per response ([0.0071, 0.0104]; paired fold SD 0.0034). Its fitted direct-30/(adjacent-15+15) compression magnitude is 0.752 ([0.751, 0.754]), against 0.757 in the 2022 report and one under additivity. The code also exactly reproduces the published 64-cell consensus regression ($\beta_{\bar{a}} = -0.00510$, $t = -5.566$, $R^2 = 0.556$) as a checksum; it is not substituted for cross-validation.

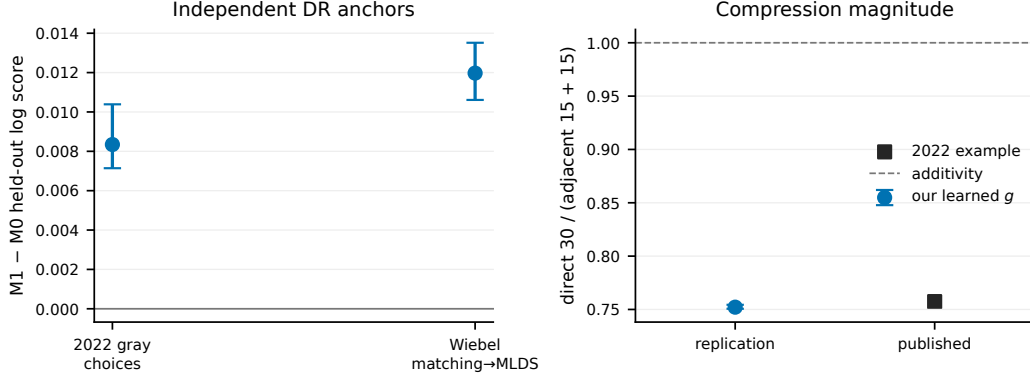


Figure 1: Gray-axis anchors. Points are out-of-fold effects and bars are paired 95% training-cluster bootstrap intervals. The right panel compares the scale-free direct-to-parts compression ratio with the 2022 example; one is additive.

Table 1: Primary results. Chromatic scores equal-weight Helm and Ekman (larger is better). Intervals are paired 95% training-cluster bootstrap intervals; fold SD is the sample SD across five held-out folds. The gray magnitude is direct 30/(adjacent 15+15), with one additive.

Quantity	Estimate	95% CI	Comparison	Notes
Gray DR replication: score; magnitude	+0.0083; 0.752	[0.0071,0.0104]; [0.751,0.754]	M1 – M0 score; 2022 magnitude 0.757	paired fold SD 0.0034; direct 30/(15+15), 1 is additive
M0 additive	-1.631	[-1.749,-1.615]	held-out log score	fold SD 0.080
M1 isotropic DR	-1.178	[-1.473,-1.121]	held-out log score	fold SD 0.224
M2 cone-axis DR	-1.213	[-1.457,-1.158]	held-out log score	fold SD 0.149
M3 opponent	-0.988	[-1.289,-0.931]	held-out log score	fold SD 0.215
M3 sRGB	-1.244	[-1.519,-1.185]	held-out log score	fold SD 0.157
M3 CIE RGB	-1.080	[-1.281,-1.017]	held-out log score	fold SD 0.178
M3 CIELAB	-1.176	[-1.455,-1.138]	held-out log score	fold SD 0.169
M3 CIE LCh/HCL	-0.955	[-1.222,-0.926]	held-out log score	fold SD 0.174
Direction dependence	-0.035	[-0.102,0.050]	M2 – M1	paired fold SD 0.143
Cone alignment (selection-aware)	-0.258	[-0.387,-0.132]	M2 minus CIE LCh/HCL	M3 reselected within each draw; paired fold SD 0.098
Identifiability: resolved granularity = coarse LM/S	-0.026; -0.287	[-0.080,0.062]; [-0.385,-0.188]	LM/S vs M1; vs coarse opponent	real organization unsupported; LM/S-gen. power 90% [85%,93%]; opponent-gen. power 100% [98%,100%]
Hue-curvature control	+0.403; +0.408	[0.312,0.470]; [0.254,0.478]	DR gain on arc; chord	compression survives chord baseline

5 Experimental Results

5.1 The gray pipeline reproduces suprathreshold compression

The held-out gray comparison favors the concave M1 model over additive M0 by +0.0083 nats per response (95% CI [0.0071,0.0104]; Figure 1). Its mean training-fold outer exponent is $q = -5.00$ ([-5.00,-5.00]), saturating the prespecified lower bound. More interpretable than that gauge-dependent exponent, a direct 30-unit interval is 0.752 ([0.751,0.754]) times the sum of its two adjacent 15-unit parts on the fitted scale. Bujack et al.’s concrete reported example gives 0.757. The independently estimated Wiebel matching scale gives the same qualitative result when predicting participant-disjoint MLDS choices: M1 versus M0 is +0.0120 ([0.0106,0.0135]), with $q = -4.04$ ([-4.58,-3.46]). Paired fold SDs are 0.0034 (gray) and 0.0085 (Wiebel); all model/fold scores are in Table 3. These replications are not pooled into the chromatic likelihood.

5.2 Isotropic compression helps; cone-axis flexibility does not

Figure 2 and Table 1 show the complete prespecified panel. Moving from additive M0 to isotropic M1 improves equal-dataset held-out log density by +0.453 ([0.246,0.524]). M2 then changes it by

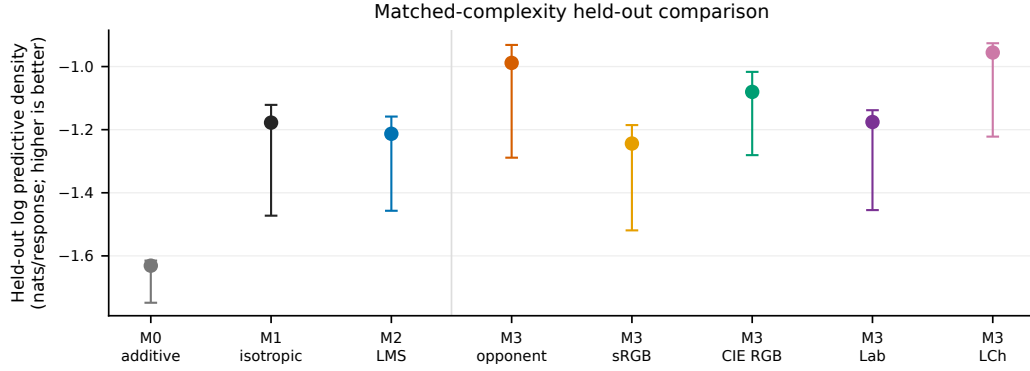


Figure 2: All primary models on the same pair-disjoint folds. Every M3 has the same parameter count and functional form as M2. Bars are paired 95% bootstrap intervals for absolute scores; headline inference uses paired score differences rather than overlap of these marginal bars.

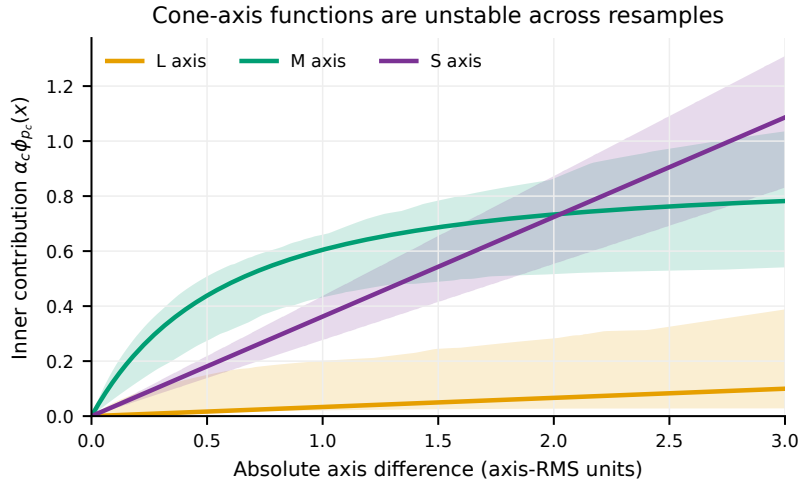


Figure 3: Fitted cone-axis inner contributions in the fixed RMS/von-Kries gauge. Lines are the full-data descriptive optimum; bands refit the corresponding training bootstrap. Distinct point curves are allowed by M2, but their instability and lack of held-out gain preclude a cone-function interpretation.

only -0.035 ($[-0.102, 0.050]$) relative to M1. Extra cone-axis curvature therefore does not earn its complexity on unseen pairs.

Model fold SDs span 0.080 to 0.224 ; paired SDs are 0.219 (M1 minus M0), 0.143 (M2 minus M1), and 0.098 (M2 minus point-best M3). Table 3 gives every fold score.

The stronger alignment test is also negative. The resample-selected contrast M2 minus the best M3 is -0.258 ($[-0.387, -0.132]$). The point winner is CIE LCh/HCL, which is selected in 75% of bootstrap draws. Selection is repeated inside every draw; the interval does not condition on that point winner. Indeed, the best M3 improves over M1 by $+0.222$ ($[0.103, 0.400]$), whereas the opponent control alone improves by $+0.189$ ($[0.014, 0.294]$). Thus the data contain predictable axis-dependent structure, but the prespecified LMS organization does not explain it.

The primary outer exponents are $q = -3.20$ ($[-4.36, -1.66]$) for M1 and $q = -1.45$ ($[-2.67, -0.14]$) for M2. M2 is distinct at its descriptive optimum: $(p_L, p_M, p_S) = (1.00, -1.77, 1.00)$ with CIs $[-3.80, 0.94]/[-3.81, 0.11]/[0.89, 1.00]$, and weights $(0.03, 0.60, 0.36)$ with CIs $[0.02, 0.20]/[0.43, 0.66]/[0.28, 0.44]$. The $L - M$, $L - S$, and $M - S$ contrast CIs are $[-2.85, 3.05]/[-4.80, -0.06]/[-4.81, -0.89]$. Bounds and poor recovery preclude physiological curve interpretation (Figure 3); Table 4 reports every primary-panel scaling parameter and CI.

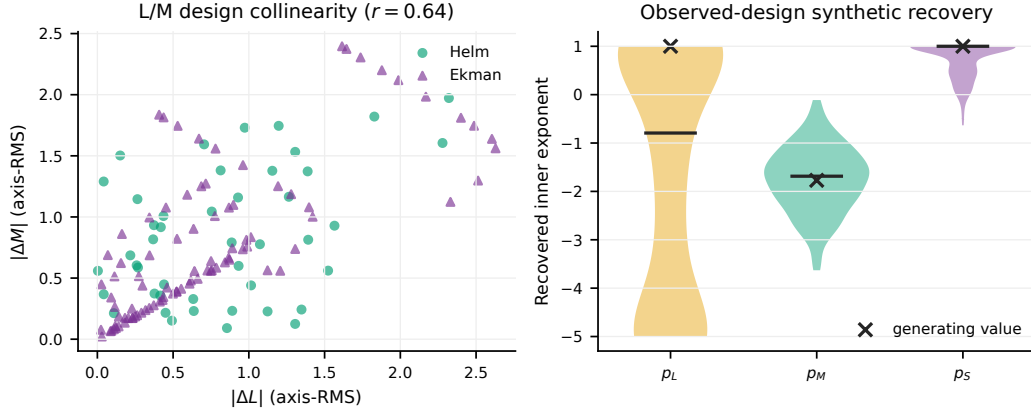


Figure 4: Identifiability. Left: observed RMS-normalized L/M pair directions. Right: full-M2 exponent recovery from 200 synthetic datasets using the observed design and fitted noise; crosses mark generating values. Model identity is highly discriminable even though p_L is not.

5.3 The null is resolvable, but individual cone shapes are not

The combined pair design is full rank: the absolute $L-M$ correlation is 0.641, with absolute and signed standardized condition numbers 2.64 and 3.82. Full rank does not imply parameter recovery. In 200 simulations from the fitted M2 on the exact observed directions, the RMSEs for p_L, p_M, p_S are 3.63, 0.69, and 0.46; their empirical 95% widths are 6.00, 2.63, and 1.11. In particular, the L exponent spans the entire allowed range.

This weak shape recovery does not make the model-comparison null uninformative. Cross-validated M2-versus-opponent selection is correct in 98% ([96%,99%]) of cone-generated simulations and 100% ([98%,100%]) of opponent-generated simulations. The exact stimulus directions can therefore discriminate the two fitted organizations even though they cannot recover every cone exponent.

At the prespecified coarser granularity, LM/S loses -0.026 ([-0.080,0.062]) to its isotropic nested model and -0.287 ([-0.385,-0.188]) to the matched two-channel opponent alternative. The same decision rule succeeds in 90% ([85%,93%]) of coarse-LMS-generated and 100% ([98%,100%]) of coarse-opponent-generated simulations. Hence neither a three-cone split nor the coarser LM/S organization is supported; the coarse null is earned at a granularity the design demonstrably resolves.

5.4 Curvature and specification controls

A curved stimulus locus can make accumulated local steps exceed an endpoint chord without perceptual compression. On the explicit polygonal exposure, concave scaling improves over linear scaling by +0.403 ([0.312,0.470]). Crucially, it also improves by +0.408 ([0.254,0.478]) when exposure is the direct ambient chord; the corresponding outer exponents are -0.65 ([-1.16,-0.36]) and -1.51 ([-2.29,-0.81]). Under linear scaling, chord minus arc is only -0.008 ([-0.090,0.033]). Curvature of the sampled polygon is therefore insufficient to explain the predictive compression.

The channel verdict is stable across the direct-chord primary analysis, periodic/open polygon exposures, response-blind smooth paths, removal of Helm C, exact-at-wavelength Ekman LMS values, and the anchored-log family (Table 2; Figure 5). The local-pair restriction is deliberately severe and its broad interval is uninformative, not contrary evidence. Color-deficient Helm sessions are reported separately and never mixed into the primary observer sample.

6 Analysis and Interpretation

Three prespecified answers. First, M2 does not beat M1, so cone-axis direction dependence is unsupported. The best non-cone M3 does beat M1: direction dependence exists but is not LMS-organized or a repackaging of prior non-constancy [Stark et al., 2025]. Second, M2 loses to the

Table 2: Prespecified chromatic sensitivities. Each cell is a held-out log-score difference with a paired 95% bootstrap interval. “Best M3” is reselected within each draw.

Specification	M2 minus M1	M2 minus best M3
Direct chord (primary)	-0.035 [-0.102,0.050]	-0.258 [-0.387,-0.132]
Polygonal path exposure	+0.007 [-0.051,0.057]	-0.374 [-0.415,-0.241]
Smooth-path exposure	+0.031 [-0.044,0.074]	-0.370 [-0.412,-0.243]
Local, low-curvature pairs	+0.069 [-0.462,0.856]	-0.002 [-2.029,0.183]
Remove Helm C	+0.035 [-0.097,0.094]	-0.452 [-0.634,-0.354]
Exact spectral Ekman LMS	-0.050 [-0.122,0.038]	-0.196 [-0.334,-0.077]
Anchored-log family	-0.028 [-0.068,0.032]	-0.277 [-0.388,-0.148]
Color-deficient Helm	+0.125 [0.002,0.242]	+0.000 [-0.373,0.035]

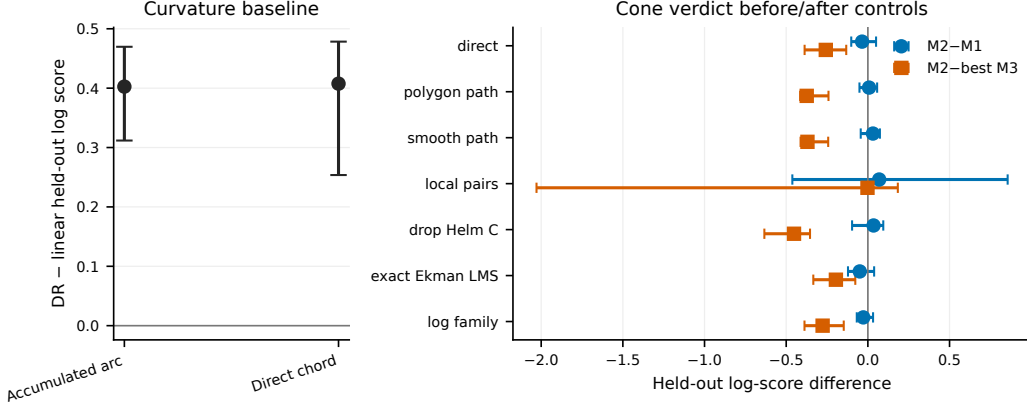


Figure 5: Curvature and specification controls. Left: concavity earns held-out score for both accumulated arc and direct-chord exposure. Right: cone-axis M2 does not improve on M1 or the best matched non-cone basis under any informative route.

resample-selected panel, rejecting cone alignment. Third, no granularity passes: exponent recovery is partial, and powered coarse LM/S loses to isotropic and opponent alternatives.

What the earned null means. The null is stronger than “too few directions”: the observed design recovers the identity of fitted LMS versus opponent generators in both directions, while real and powered coarse tests fail. The compact pre-combination model is therefore unsupported. All coordinates derive from cones, but L , M , S separability is not predictive here; LCh/HCL is a coordinate system, not a neural channel, and opponent improvement identifies no circuit. The data do not localize compression before cone combination.

DR versus locus geometry. Compression improves direct-chord as well as accumulated-path prediction, and the cone verdict persists under polygonal and smooth exposures. Sampled-locus curvature is insufficient, although pairs do not reveal the perceptual geodesic. This is not a new non-Riemannian proof; the ordered neutral locus retains that stronger conditional reading.

Failure analysis and limits. Although 99.9% of fits converge, 82.1% hit a bound and p_L spans its full range; fine shapes and local-pair effects are unidentified. Approximate colorimetry and missing participant metadata limit physiological and population inference (Appendix C). A decisive study must separate measured L , M , S directions and hold out observers and directions.

7 Conclusion

Testing the per-cone proposal rather than anisotropy alone, we find that isotropic compression beats additivity but separate LMS functions and a powered coarse LM/S split fail. This earned null does not support a compact pre-combination mechanism. Future alignment claims need matched controls, fixed gauge, curvature checks, and recovery on designs separating L from M .

References

- Roxana Bujack, Emily Teti, Jonah Miller, Elektra Caffrey, and Terece L. Turton. The non-Riemannian nature of perceptual color space. *Proceedings of the National Academy of Sciences*, 119(18): e2119753119, 2022. doi: 10.1073/pnas.2119753119. URL <https://doi.org/10.1073/pnas.2119753119>.
- Roxana Bujack, Emily N. Stark, Terece L. Turton, Jonah Maxwell Miller, and David H. Rogers. The geometry of color in the light of a non-Riemannian space. *Computer Graphics Forum*, 44(3): e70136, 2025. ISSN 1467-8659. doi: 10.1111/cgf.70136. URL <https://doi.org/10.1111/cgf.70136>.
- Commission Internationale de l’Eclairage. CIE 160:2004: A review of chromatic adaptation transforms. Technical Report CIE 160:2004, Commission Internationale de l’Eclairage, Vienna, Austria, 2004. URL <https://www.cie.co.at/publications/review-chromatic-adaptation-transforms>.
- Commission Internationale de l’Eclairage. CIE 2006 LMS cone fundamentals for 2 degree field size in terms of energy. CIE data set, 2006. URL <https://doi.org/10.25039/CIE.DS.tijidesg>. Data table CIE_lms_cf_2deg.csv; source: CIE 170-1:2006, Table 6.7; CIE copyright and terms apply.
- Commission Internationale de l’Eclairage. CIE 015:2018: Colorimetry, 4th edition. Technical Report CIE 015:2018, Commission Internationale de l’Eclairage, Vienna, Austria, 2018. URL <https://doi.org/10.25039/TR.015.2018>.
- Commission Internationale de l’Eclairage. CIE 1931 colour-matching functions, 2 degree observer. CIE data set, 2019. URL <https://doi.org/10.25039/CIE.DS.xvudnb9b>. Data table CIE_xyz_1931_2deg.csv; CIE copyright and terms apply.
- Gösta Ekman. Dimensions of color vision. *The Journal of Psychology*, 38(2):467–474, 1954. doi: 10.1080/00223980.1954.9712953. URL <https://doi.org/10.1080/00223980.1954.9712953>.
- Carl E. Helm. Multidimensional ratio scaling analysis of perceived color relations. *Journal of the Optical Society of America*, 54(2):256–262, 1964. doi: 10.1364/JOSA.54.000256. URL <https://doi.org/10.1364/JOSA.54.000256>.
- International Electrotechnical Commission. IEC 61966-2-1:1999: Multimedia systems and equipment – colour measurement and management – part 2-1: Colour management – default RGB colour space – sRGB. Technical Report IEC 61966-2-1:1999, International Electrotechnical Commission, 1999. URL <https://webstore.iec.ch/en/publication/6169>.
- International Organization for Standardization and Commission Internationale de l’Eclairage. ISO/CIE 11664-4:2019(E): Colorimetry – part 4: CIE 1976 L*a*b* colour space. Technical Report ISO/CIE 11664-4:2019(E), ISO and CIE, 2019. URL <https://www.cie.co.at/publications/colorimetry-part-4-cie-1976-lab-colour-space-1>.
- International Organization for Standardization and Commission Internationale de l’Eclairage. ISO/CIE 11664-6:2022(E): Colorimetry – part 6: CIEDE2000 colour-difference formula. Technical Report ISO/CIE 11664-6:2022(E), ISO and CIE, 2022. URL <https://www.cie.co.at/publications/colorimetry-part-6-ciede2000-colour-difference-formula-1>.
- Los Alamos National Laboratory. color: Studies on color, colormaps, and colorspace. Source-code and data repository, 2022. URL <https://github.com/lanl/color>. Revision 099491a0737248ffed75524cf01c66289bf9e3d; BSD 3-Clause License; LANL release C20137.
- M. Ronnier Luo, Guihua Cui, and Bryan Rigg. The development of the CIE 2000 colour-difference formula: CIEDE2000. *Color Research & Application*, 26(5):340–350, 2001. doi: 10.1002/col.1049. URL <https://doi.org/10.1002/col.1049>.
- Donald I. A. MacLeod and Robert M. Boynton. Chromaticity diagram showing cone excitation by stimuli of equal luminance. *Journal of the Optical Society of America*, 69(8):1183–1186, 1979. doi: 10.1364/JOSA.69.001183. URL <https://doi.org/10.1364/JOSA.69.001183>.

- Patrick Mair, Patrick J. F. Groenen, and Jan de Leeuw. More on multidimensional scaling and unfolding in R: smacof version 2. *Journal of Statistical Software*, 102(10):1–47, 2022. doi: 10.18637/jss.v102.i10. URL <https://doi.org/10.18637/jss.v102.i10>.
- Laurence T. Maloney and Joong Nam Yang. Maximum likelihood difference scaling. *Journal of Vision*, 3(8):573–585, 2003. doi: 10.1167/3.8.5. URL <https://doi.org/10.1167/3.8.5>.
- Emily Stark, Terece L. Turton, and Roxana Bujack. Diminishing returns in perceptual color space – now in color. In Mennatallah El-Assady, Alvitta Ottley, and Christian Tominski, editors, *EuroVis 2025 – Short Papers*. The Eurographics Association, 2025. ISBN 978-3-03868-282-0. doi: 10.2312/evs.20251081. URL <https://doi.org/10.2312/evs.20251081>.
- Emily S. Teti, Terece L. Turton, Jonah M. Miller, and Roxana Bujack. Maximum likelihood estimation of difference scaling functions for suprathreshold judgments. *Journal of Vision*, 22(10):9, 2022. doi: 10.1167/jov.22.10.9. URL <https://doi.org/10.1167/jov.22.10.9>.
- L. L. Thurstone. A law of comparative judgment. *Psychological Review*, 34(4):273–286, 1927. doi: 10.1037/h0070288. URL <https://doi.org/10.1037/h0070288>.
- Christiane B. Wiebel, Guillermo Aguilar, and Marianne Maertens. Maximum likelihood difference scales represent perceptual magnitudes and predict appearance matches. *Journal of Vision*, 17(4): 1–14, 2017. doi: 10.1167/17.4.1. URL <https://doi.org/10.1167/17.4.1>.
- Christiane B. Wiebel, Guillermo Aguilar, and Marianne Maertens. Online material for “maximum likelihood difference scales represent perceptual magnitudes and predict appearance matches”. Open Science Framework, 2022. URL <https://osf.io/58jz3/>. Archived data and analysis materials for the 2017 article. The README supplied with this benchmark states a Creative Commons Attribution 4.0 International license.

A Conversion, Conditioning, and Verification

The least-squares bridge from the supplied CIE 1931 color-matching functions to the supplied CIE 2006 energy-normalized fundamentals is

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} = \begin{bmatrix} 0.2682740 & 0.8466136 & -0.0348920 \\ -0.3859447 & 1.1648773 & 0.1028512 \\ 0.0212467 & -0.0245520 & 0.5341609 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}.$$

The spectral projection has channelwise R^2 values 0.9992, 0.9992, and 0.9781. After componentwise von-Kries normalization, the declared C white maps to (1, 1, 1) by algebra; the independent Helm-A check has $L = 0.338$. At Ekman 555 nm, exact fundamental lookup and projected XYZ differ by 0.0060 in normalized S , illustrating why the exact-spectral sensitivity is necessary. The machine-readable logs contain the complete matrix, inverse, residual diagnostics, negative-prediction counts, white values, axis RMS scales, hashes, and all converted coordinates.

Axis RMS conditioning is calculated once from the stimulus pairs and never from judgments. It makes a unit coordinate difference comparable across bases while preserving each fixed axis direction. It is not covariance whitening: rotating to sample principal components would create a design-adaptive M3 and invalidate the alignment test.

B Additional Held-out Results

The M2 minus M1 effects split by human dataset are +0.037 ([-0.059,0.148]) for Helm and -0.107 ([-0.227,0.091]) for Ekman. Their disagreement reinforces the need for equal-dataset scoring and cluster uncertainty rather than treating Helm’s repeated sessions as independent directions.

The power-family HCL and opponent gains over M1 are respectively +0.222 ([0.045,0.400]) and +0.189 ([0.014,0.294]). These are secondary paired contrasts; the primary cone interval remains the more conservative comparison with the best control reselected inside every bootstrap draw.

Table 3: All point-analysis held-out fold scores and their sample SD. The M2 versus M3 row conditions on the full-sample point winner (LCh/HCL); selection-aware uncertainty is supplied separately by the bootstrap contrast.

Model/comparison	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	SD
<i>Chromatic equal-dataset log score</i>						
M0	-1.602	-1.675	-1.611	-1.741	-1.529	0.080
M1	-1.258	-0.954	-0.951	-1.474	-1.251	0.224
M2	-1.218	-1.237	-0.987	-1.407	-1.219	0.149
M3 opponent	-0.959	-0.871	-0.779	-1.345	-0.992	0.215
M3 sRGB	-1.368	-1.275	-0.987	-1.367	-1.222	0.157
M3 CIE RGB	-1.130	-1.033	-0.826	-1.320	-1.093	0.178
M3 CIELAB	-1.111	-1.238	-0.970	-1.426	-1.138	0.169
M3 LCh/HCL	-0.862	-1.138	-0.744	-1.133	-0.904	0.174
<i>Paired chromatic differences</i>						
M1 minus M0	0.344	0.721	0.661	0.267	0.278	0.219
M2 minus M1	0.041	-0.283	-0.037	0.067	0.032	0.143
M2 minus point-best M3	-0.355	-0.099	-0.243	-0.274	-0.315	0.098
<i>Gray/Wiebel log score and paired differences</i>						
Gray M0	-0.6452	-0.6529	-0.6372	-0.6281	-0.6531	0.0107
Gray M1	-0.6330	-0.6413	-0.6310	-0.6213	-0.6484	0.0103
Gray M1 minus M0	0.0122	0.0117	0.0062	0.0068	0.0047	0.0034
Wiebel M0	-0.5755	-0.5592	-0.5084	-0.5101	-0.4518	0.0487
Wiebel M1	-0.5694	-0.5380	-0.4891	-0.5089	-0.4397	0.0492
Wiebel M1 minus M0	0.0061	0.0212	0.0193	0.0012	0.0121	0.0085

Table 4: Primary full-data scaling parameters with 95% training-cluster bootstrap percentile intervals. Each weight and inner-exponent line follows the three-axis order defined in Section 3.3; fixed values are not estimated. Point estimates can fall outside percentile intervals when full-data and bootstrap training designs differ.

Model/basis	Outer q	Weights $\alpha_1, \alpha_2, \alpha_3$	Inner p_1, p_2, p_3
M0 (LMS)	fixed 1	0.19 [0.12,0.25]	fixed (1, 1, 1)
		0.34 [0.25,0.41]	
		0.47 [0.43,0.52]	
		0.01 [0.01,0.22]	
		0.64 [0.40,0.69]	
M1 (LMS)	-3.20 [-4.36,-1.66]	0.36 [0.26,0.47]	fixed (1, 1, 1)
		0.03 [0.02,0.20]	1.00 [-3.80,0.94]
		0.60 [0.43,0.66]	-1.77 [-3.81,0.11]
M2 (LMS)	-1.45 [-2.67,-0.14]	0.36 [0.28,0.44]	1.00 [0.89,1.00]
		0.59 [0.44,0.67]	1.00 [-0.71,1.00]
		0.31 [0.23,0.41]	1.00 [0.91,1.00]
M3 opponent	-2.51 [-2.84,-0.95]	0.09 [0.03,0.24]	-5.00 [-5.00,-2.60]
		0.38 [0.28,0.48]	1.00 [-2.08,1.00]
		0.20 [0.11,0.31]	-5.00 [-5.00,-0.20]
M3 sRGB	-1.51 [-2.56,-0.23]	0.42 [0.36,0.47]	1.00 [1.00,1.00]
		0.18 [0.07,0.40]	-5.00 [-3.70,1.00]
		0.40 [0.20,0.52]	1.00 [-3.33,0.61]
M3 CIE RGB	-1.14 [-2.11,-0.44]	0.41 [0.34,0.46]	1.00 [0.93,1.00]
		0.02 [0.01,0.18]	-5.00 [-4.86,-0.83]
		0.53 [0.41,0.56]	0.16 [-2.02,0.85]
M3 CIELAB	-2.07 [-2.52,-0.88]	0.44 [0.36,0.51]	1.00 [0.78,1.00]
		0.04 [0.02,0.17]	-5.00 [-5.00,-1.66]
		0.00 [0.00,0.07]	-5.00 [-5.00,-3.88]
M3 LCh/HCL	-0.22 [-1.25,0.60]	0.96 [0.81,0.96]	-0.78 [-1.56,0.59]

C Audit and Reproducibility

All analyses use seed 20260728. Raw bootstrap rows store their training clusters, fold predictions, parameters, optimizer status, and bounds. Raw synthetic rows are explicitly marked by generator and are never included as human observations. The analysis-plan timestamp predates fitting; the attempts log records data defects, implementation smoke failures, and every deviation forced by the supplied files. The convergence audit reports every status and retains the prespecified 200 paired draws rather than deleting unfavorable or boundary solutions. In the primary panel 99.9% of 8,000 selected bootstrap-fold fits converge; across all eight panels the corresponding rate is 99.9%.

Extended failure analysis. Ekman’s spectral directions alone are signed-rank deficient. Full-design recovery fits converge in 98.5% of draws but hit a bound in 82.5%; across chromatic panels the boundary rate is 82.1%. The gray outer exponent also saturates its lower bound. Fine curve shapes are therefore not interpreted, even when percentile contrast intervals are asymmetric. The local-pair restriction is underpowered, whereas path, drop-C, exact-spectral, and family controls agree on the primary alignment sign.

The missing advertised cone transform forces a logged least-squares bridge, whose S projection is least accurate. Exact Helm excitations cannot be reconstructed without reflectances and an Illuminant-C spectrum. More generally, Gaussian links, fixed bases, L_1 aggregation, bounded curvature families, and dataset-specific RMS conditioning delimit the comparison. Training-cluster intervals cannot replace population sampling when Ekman is aggregate and gray participant identifiers are absent.

From the workspace root, the computational sequence is:

```
PYTHONPATH=proposal/vendor:proposal/code python3 proposal/code/cone_space.py
PYTHONPATH=proposal/vendor:proposal/code python3 proposal/code/fit_gray.py
PYTHONPATH=proposal/vendor:proposal/code python3 proposal/code/fit_chromatic.py
PYTHONPATH=proposal/vendor:proposal/code python3 proposal/code/confound.py
PYTHONPATH=proposal/vendor:proposal/code python3 proposal/code/identifiability.py
PYTHONPATH=proposal/vendor:proposal/code python3 proposal/code/audit_results.py
PYTHONPATH=proposal/vendor:proposal/code python3 proposal/code/make_numbers.py
PYTHONPATH=proposal/vendor:proposal/code python3 proposal/code/make_figures.py
```

The paper is then compiled from proposal/ with pdflatex, bibtex, and two further pdflatex passes.

D Provenance and Ethical Scope

No new participants were recruited and no external human color data were used. The LANL gray release is BSD-3-Clause; the supplied Wiebel archive is CC BY 4.0; Helm and Ekman matrices arrive through the GPL-3 smacof distribution with original attribution. CIE/CVRL tabulations are cited as scientific data rather than represented as permissively licensed software. The analysis uses historical and aggregate judgments solely for perceptual model comparison; it makes no decisions about individuals.